

Dr. Doc:

 2014 Exam 

The "pseudostratified ciliated columnar epithelium with goblet cells that lines the trachea is embryological derived from:

- A. Neuroectoderm
- B. Endoderm
- C. Ectoderm
- D. Visceral mesoderm

The correct answer is B. Endoderm.

To determine the germ layer origin of any part of the respiratory system, remember the basic rule of respiratory embryology:

1. Endoderm Origin (Option B)

- The entire inner epithelial lining of the respiratory tract—from the larynx, trachea, and bronchi down to the terminal alveoli—as well as the respiratory glands (and goblet cells), is derived directly from the endoderm of the primitive foregut (specifically via the respiratory diverticulum or lung bud).

2. Visceral (Splanchnic) Mesoderm Origin (Option D)

- The surrounding structural components of the trachea and lungs—including the cartilage rings, smooth muscle, connective tissue, and blood vessels—are derived from the visceral (splanchnic) mesoderm that surrounds the foregut.

In which stage of lung maturation does "the blood-air barrier" fully established?

- A. Pseudoglandular
- B. Canalicular
- C. Terminal sac
- D. Alveolar

The correct answer is C. Terminal sac (also known as the Saccular stage).

The establishment of the blood-air barrier (the ultra-thin membrane consisting of Type I pneumocytes, fused basal laminae, and capillary endothelial cells) is the critical milestone that makes extrauterine survival possible.

Let's look at the development of this barrier across the four major stages of lung maturation:

1. Pseudoglandular Stage (Weeks 5–16)

- No respiratory bronchioles or alveoli exist yet.
- Resembles an exocrine gland. Survival is impossible.

2. Canalicular Stage (Weeks 16–26)

- Respiratory bronchioles and alveolar ducts form.
- Capillaries begin to establish contact with the developing airways, but the barrier is still relatively thick.

- Toward the very end of this stage (around week 24–26), early Type II pneumocytes begin producing small amounts of surfactant, but the barrier is not fully established.

3. Terminal Sac / Saccular Stage (Weeks 26 – Birth) — CORRECT ANSWER

- Terminal sacs (primitive alveoli) develop in large numbers.
- The epithelium thins dramatically into flat Type I pneumocytes.
- Capillaries bulge into these primitive alveoli, and their basement membranes fuse with those of the Type I pneumocytes.
- This creates a functional, fully established, ultra-thin blood-air barrier, allowing adequate gas exchange for a premature infant to survive.

4. Alveolar Stage (32 Weeks – 8 Years Postnatal)

- Mature alveoli with hyper-developed capillary networks form. (While gas exchange expands rapidly here, the barrier itself was already fully established in the terminal sac stage to allow breathing at birth).

In the development of respiratory system, the cranial portion of the foregut is divided into ventral part and dorsal part by the tracheoesophageal. Which one of the following anatomical structure is developed from the dorsal part?

- A. Oropharynx
- B. Trachea
- C. Bronchi
- D. lungs

The correct answer is A. Oropharynx (or Esophagus / GI tract).

During the 4th week of embryonic development, the respiratory diverticulum (lung bud) appears as an outgrowth on the ventral wall of the primitive foregut.

To separate the respiratory system from the digestive system, a pair of longitudinal ridges called the tracheoesophageal folds fuse in the midline to form the tracheoesophageal septum.

This septum divides the cranial portion of the foregut into two distinct compartments:

1. The Ventral Part (Respiratory Tube)

Gives rise to the:

- Trachea (Option B)
- Bronchi (Option C)
- Lungs / Respiratory tree (Option D)
- Larynx (infraglottic portion)

2. The Dorsal Part (Digestive Tube)

Gives rise to the:

- Pharynx / Oropharynx (Option A)

- Esophagus

Clinical Correlation: Tracheoesophageal Fistula (TEF)

If the tracheoesophageal septum fails to deviate or fuse properly, it results in an abnormal persistent connection between the ventral respiratory tube and the dorsal digestive tube. This causes a Tracheoesophageal Fistula (TEF), classically associated with esophageal atresia, polyhydramnios in utero, and immediate choking/cyanosis upon the newborn's first feeding.

Epiglottis develops from;

- A. Caudal part of the hypopharyngeal eminence
- B. Respiratory diverticulum
- C. Mesenchyme of 4th and 6th pharyngeal arches
- D. Cranial-most part of laryngotracheal tube

During the fusion of tracheoesophageal folds, small opening remain connecting the laryngotracheal tube into the pharynx and soon forms;

- A. Primordial laryngeal Inlet
- B. Primordial pharyngeal Inlet
- C. Epiglottis
- D. Vocal folds

The correct answer is A. Primordial laryngeal Inlet.

During the 4th and 5th weeks of embryonic development, the tracheoesophageal septum separates the ventral respiratory tube (laryngotracheal tube) from the dorsal digestive tube (esophagus).

1. Formation of the Primordial Laryngeal Inlet

- As the tracheoesophageal folds fuse in a caudal-to-cranial direction, the division of the foregut into two tubes is completed along the length of the future trachea.

- However, at the most cranial end, the fusion stops. This leaves an opening through which the newly formed laryngotracheal tube maintains communication with the primitive pharynx.

- This persistent cranial opening becomes the primordial laryngeal inlet (or laryngeal orifice).

2. Subsequent Remodeling into the Adult Inlet

- Initially, this inlet is a simple, longitudinal T-shaped slit bounded by the epiglottic swelling (cranially) and two arytenoid swellings (caudally/laterally).

- Later, the internal laryngeal epithelium proliferates and temporarily occludes the lumen. Recanalization occurs by the 10th week, shaping the laryngeal inlet, false vocal cords, true vocal cords, and laryngeal ventricles.

In development, during what timeframe does a fetus have enough mature structures to breathe on own if born prematurely?

A. Week 28

B. Week 36

C. Week 16

D. Week 21

The correct answer is A. Week 28.

Extrauterine survival for a premature fetus depends on two absolute physiological prerequisites:

1. A fully established Blood-Air Barrier: Ultra-thin Type I pneumocytes in direct contact with capillaries to allow gas exchange.
2. Sufficient Pulmonary Surfactant: Secreted by Type II pneumocytes to prevent alveolar collapse during expiration.

Let's look at the timeline to see why Week 28 is the classic textbook milestone:

1. Weeks 16 – 21 (Canalicular Stage - Options C & D)

- Status: Respiratory bronchioles are just beginning to form.
- Surfactant production has not yet begun (it begins around Week 20–24), and the capillary-airway distance is far too thick. Survival is impossible.

2. Week 26 to Week 28 (Terminal Sac Stage - OPTION A)

- The Critical Milestone: By Week 26 to 28, a sufficient number of terminal sacs (primitive alveoli) have developed, lined by thin Type I pneumocytes with an established capillary network (the blood-air barrier).

- Crucially, Type II pneumocytes have matured enough to secrete functional amounts of pulmonary surfactant.

- Therefore, by Week 28, a prematurely born fetus has enough mature respiratory structures and surfactant to breathe on its own (though intensive care support may still be required).

3. Week 36 (Alveolar Stage - Option B)

- By Week 36, surfactant production is fully abundant. While a fetus born at Week 36 breathes easily, Week 28 is the earliest established threshold where independent extrauterine breathing becomes viable.

High-Yield Board Milestone:

- Week 20–24: Surfactant production begins.
- Week 26–28: Earliest viability threshold (Enough mature terminal sacs + surfactant to breathe on its own).
- Week 35–36: Surfactant reaches fully mature levels.

During the 4th week of development, the incomplete division of the cranial part of the foregut into respiratory and esophageal parts results in the;

A. Laryngeal atresia

B. Tracheoesophageal fistula

C. esophageal atresia

D. Lung agenesis

Correct Answer: B. Tracheoesophageal fistula (TEF)

Smooth muscle, connective tissue, and cartilage of primary bronchi are derived from which of the following sources?

A. Neuroectoderm

B. Endoderm

C. Ectoderm

D. Visceral mesoderm

Correct Answer: D. Visceral mesoderm

Which is the earliest period of lung maturation to support respiration in a premature fetus?

A. Alveolar

B. Terminal saccular

C. Pseudoglandular

D. Canalicular

The correct answer is B. Terminal saccular (Terminal sac stage).

This question asks for the earliest stage of lung maturation during which extrauterine gas exchange (respiration) becomes physically possible for a premature fetus.

Let's review the four classic stages of lung maturation:

1. Pseudoglandular Period (Weeks 5–16)

- The lung resembles an exocrine gland.
- Only conducting airways are present; no respiratory structures exist. Survival is completely impossible.

2. Canalicular Period (Weeks 16–26)

- Lumina of bronchi and terminal bronchioles enlarge; respiratory bronchioles and alveolar ducts form.
- While vascularization increases, primitive alveoli have not yet formed, and the blood-air barrier is not fully established. Survival at the end of this stage is extremely rare and difficult.

3. Terminal Saccular Period (Weeks 26 – Birth) — CORRECT ANSWER

- The Milestone Stage: Terminal sacs (primitive alveoli) develop in large numbers.
- The epithelium thins into Type I pneumocytes, forming the functional blood-air barrier.
- Type II pneumocytes begin secreting adequate amounts of pulmonary surfactant to keep the sacs open.
- This is the earliest period that can support respiration in a premature fetus.

4. Alveolar Period (32 Weeks – 8 Years Postnatal)

- True, mature alveoli develop. While respiration is fully efficient here, the earliest stage that supports respiration is the Terminal Saccular period.

 2016 Exam 

Lung maturation characterized by formation of respiratory bronchioles and few terminal sacs while fetus is unable to survive with intensive care manifested during the;

- A. Pseudoglandular period
- B. Canalicular period
- C. Terminal sac period
- D. Alveolar period
- E. Glandular period

The correct answer is B. Canalicular period.

Let's break down the key features provided in the question:

1. "Formation of respiratory bronchioles and few terminal sacs": During the canalicular period (Weeks 16–26), the terminal bronchioles divide into two or more respiratory bronchioles, which then divide into 3 to 6 alveolar ducts that begin to form early, primitive terminal sacs toward the very end of this period.
2. "Fetus is unable to survive (or rarely survives even with intensive care)": Because this stage occurs between weeks 16 and 26, the capillary-airway distance is still relatively thick, and Type II pneumocytes have not yet secreted sufficient amounts of surfactant. Fetuses born during the early/mid canalicular period usually cannot survive because the respiratory system is still too immature.

Why the other options are incorrect:

- A & E. Pseudoglandular / Glandular period (Weeks 5–16): Characterized by conducting airways only (up to terminal bronchioles). NO respiratory bronchioles or terminal sacs exist yet.
- C. Terminal sac period (Weeks 26 – Birth): Characterized by numerous terminal sacs, an established blood-air barrier, and adequate surfactant, allowing premature infants TO survive with intensive care.
- D. Alveolar period (32 Weeks – 8 Years): Characterized by mature alveoli; infants born here survive easily.

In the development of the lung, which of the following is NOT correct:

- A. Alveolar collapse as a result of insufficient surfactant production
- B. Alveoli continue to form postnatally
- C. Alveoli contain specialized epithelial cells which produce surfactant
- D. Alveolar epithelium is derived from splanchnic mesoderm
- E. None of the above

Correct Answer: D. Alveolar epithelium is derived from splanchnic mesoderm

Why?

The alveolar epithelium is derived from endoderm, specifically the endoderm of the laryngotracheal respiratory diverticulum.

The stage of lung development, during which the respiratory epithelium becomes squamous and the capillary loops are intimately related to the epithelium is the:

- A. Glandular period
- B. Pseudoglandular period

- C. Alveolar period
- D. Canalicular period
- E. None of the above

E. None of the above

▼ Why?

These features are most characteristic of the terminal sac (saccular) period, rather than the canalicular period:

Terminal sac period ($\approx$ 24 weeks–birth) $\rightarrow$ terminal sacs develop, epithelium becomes very thin/squamous, and capillaries become intimately related to the epithelial cells $\rightarrow$ formation of the blood–air barrier.

Canalicular period ($\approx$ 16–25 weeks) $\rightarrow$ respiratory bronchioles develop and vascularization increases, but the definitive intimate capillary–epithelial relationship is emphasized in the terminal sac stage.

Since “Terminal sac period” is not among the options, the best answer is None of the above

Pulmonary surfactant most likely begins to form in the human fetus at about;

- A. 16 weeks
- B. 20 weeks
- C. 28 weeks
- D. 32 weeks

The correct answer is B. 20 weeks (or 20–24 weeks).

Understanding the timeline of pulmonary surfactant production

1. Start of Production (~Week 20):

- During the Canalicular stage (Weeks 16–26), specialized Type II alveolar epithelial cells (Type II pneumocytes) begin to differentiate.
- By approximately 20 weeks of gestation, these cells begin to synthesize and secrete low levels of surfactant (a mixture of phospholipids, primarily dipalmitoylphosphatidylcholine [DPPC], and proteins) into the developing airways.

2. Functional Levels (~Week 26–28):

- By 26 to 28 weeks (Terminal Sac stage), surfactant levels become high enough to allow premature infants a chance at extrauterine survival.

3. Mature Levels (~Week 35–36):

- Surfactant production ramps up dramatically during the late stages of pregnancy, reaching fully mature levels by around 35 to 36 weeks.

The terminal sac or saccular stage of lung development is characterized by the appearance of:

- A. Secondary bronchi
- B. Tertiary bronchi
- C. Respiratory bronchi
- D. Surfactant
- E. Mature alveoli

The correct answer is D. Surfactant (specifically the appearance of significant, functional surfactant secretion into primitive alveoli/sacs).

Let's look at when each listed structure first appears across the stages of lung development:

- A. Secondary bronchi & B. Tertiary bronchi: These appear very early, during the Embryonic period (Weeks 4–7) and early Pseudoglandular stage (Weeks 5–16).
- C. Respiratory bronchioles: These first appear during the Canalicular stage (Weeks 16–26) when the terminal bronchioles divide.
- E. Mature alveoli: These appear during the Alveolar stage (32 Weeks to 8 Years Postnatal).

Why D. Surfactant is the correct choice for the Saccular Stage:

- While Type II pneumocytes begin producing tiny amounts of surfactant late in the canalicular stage (~Week 20–24), it is during the Terminal Sac / Saccular stage (Weeks 26 to Birth) that surfactant secretion becomes prominent, functional, and clinically effective.
- The appearance of functional surfactant lining the newly formed terminal sacs is the defining physiological hallmark of the saccular stage, allowing the sacs to stay inflated during extrauterine breathing.

The tracheoesophageal septum separates the:

- A. Laryngotracheal tube and nasopharynx
- B. Esophagus and nasopharynx
- C. Laryngotracheal tube and esophagus
- D. Laryngotracheal tube and oropharynx
- E. Esophagus and oropharynx

Correct Answer: C. Laryngotracheal tube and esophagus

The laryngotracheal diverticulum develops within the floor of the pharynx between arches:

- A. I & II
- B. II & III
- C. III & IV
- D. IV & VI
- E. V & VI

Correct Answer: D. IV & VI

Explanation

✓The laryngotracheal diverticulum (respiratory diverticulum) appears during the 4th week as an outgrowth from the ventral wall of the caudal foregut.

✓Its cranial end is located in the floor of the developing pharynx, in the region between the:

- ▲ 4th pharyngeal arch
- ▲ 6th pharyngeal arch

✓This region later contributes to the development of the laryngeal inlet and laryngeal structures.

In which developmental stage of lung tissue, the simple cuboidal epithelium starts to differentiate into type 1 and type 2 pneumocytes?

- A. Pseudoglandular stage
- B. Canalicular stage
- C. Terminal sac
- D. Alveolar stage

The correct answer is B. Canalicular stage.

The differentiation of the respiratory epithelium into specialized cell types is a major hallmark of the Canalicular stage (Weeks 16–26):

1. Pseudoglandular Stage (Weeks 5–16):

- The airways are lined by uniform, undifferentiated simple cuboidal/columnar epithelium. No functional pneumocytes exist.

2. Canalicular Stage (Weeks 16–26)

- The simple cuboidal epithelium lining the newly formed respiratory bronchioles and alveolar ducts begins to differentiate into two distinct cell types:
- Type I Pneumocytes: Extremely thin, flat, simple squamous cells that form the primary surface area for gas exchange.
- Type II Pneumocytes: Secretory, cuboidal cells containing lamellar bodies that begin synthesizing and secreting pulmonary surfactant.

3. Terminal Sac Stage (Weeks 26 – Birth):

- The already differentiated Type I cells thin further to establish the blood-air barrier, and Type II cells ramp up surfactant secretion into the newly formed terminal sacs.

Surfactant production begins during the saccular stage of lung development.

- A. True
- B. False

Correct Answer: A. True

The tracheal epithelium is developed from which germ layer?

- A. Ectoderm
- B. Mesoderm
- C. Pharyngeal arches
- D. Endoderm

Correct Answer: D. Endoderm

Which of the following embryonic structures is responsible for separating the respiratory diverticulum with primitive foregut?

- A. Laryngotracheal groove
- B. Posterior esophageal folds
- C. Laryngotracheal diverticulum
- D. Tracheoesophageal septum

The correct answer is D. Tracheoesophageal septum.

Recanalization of the larynx, which restores the laryngeal lumen, occurs approximately at which week of embryonic development?

- A. 4th week
- B. 6th week
- C. 10th week
- D. 16th week

C. 10th week

- Explanation: After the formation of the primordial laryngeal inlet, the laryngeal lumen is temporarily occluded by a proliferation of its epithelial lining during the 5th and 6th weeks. Recanalization (the re-opening of this lumen) occurs around the 10th week of embryonic development, restoring the laryngeal airway.

In which stage of lung maturation does the lung become highly vascularized and the blood-air barrier begin to form?

- A. Pseudoglandular stage
- B. Canalicular stage
- C. Saccular stage
- D. Alveolar stage

B. Canalicular stage

- Explanation: During the canalicular stage (Weeks 16-26), the lung tissue becomes richly vascularized, and the respiratory epithelium begins to thin, making the capillaries come into intimate contact with the developing airways. This is the stage where the blood-air barrier begins to form, although it is not fully established or functional for gas exchange until the subsequent saccular stage.

In which stage of lung maturation do respiratory bronchioles first develop?

- A. Pseudoglandular stage
- B. Canalicular stage
- C. Saccular stage
- D. Alveolar stage

B. Canalicular stage

- Explanation: The respiratory bronchioles (the first part of the respiratory zone containing some alveoli) differentiate and first appear during the canalicular stage (Weeks 16-26) as the terminal bronchioles divide.

In which stage of lung maturation does the fetus have a chance of survival if born prematurely, but typically develops respiratory distress due to borderline surfactant levels?

- A. Pseudoglandular stage
- B. Canalicular stage
- C. Saccular stage
- D. Alveolar stage

C. Saccular stage

- Explanation: The saccular (terminal sac) stage (Weeks 26 to birth) is the earliest period where a premature fetus has a chance of survival. This is because the blood-air barrier is established, and Type II pneumocytes produce surfactant. However, surfactant levels are often still borderline, leading to a high risk of Respiratory Distress Syndrome (RDS).

Correction

Correct answer: A. Caudal part of the hypopharyngeal eminence

- ✓ The epiglottis develops from the caudal part of the hypopharyngeal (hypobranchial) eminence.

- ✓ The hypopharyngeal eminence is mainly derived from mesenchyme of the 3rd and 4th pharyngeal arches, with the epiglottic region particularly associated with the 4th arch.